Labour Supply 2: Labour Supply and Public Policy

EC306 - Wages and Employment

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Goals of This Section

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Today we will:

Income Maintenance Programs

Overview of Income Maintenance Programs

Government Transfers in Canada

In 2017, **83% of families** received some form of government transfer:

Program	% Receiving	Average Amount
Federal Child Benefits	20.2%	$6,314
Employment Insurance	14.0%	$8,352
Social Assistance	11.3%	$8,367
Old Age Security	25.1%	$8,566
CPP/QPP	32.4%	$10,092

Issues in Program Design

Universal vs. Targeted Programs

- Universal: easy to implement but expensive
- Targeted: benefits only those in need but costly to administer

Work Disincentives

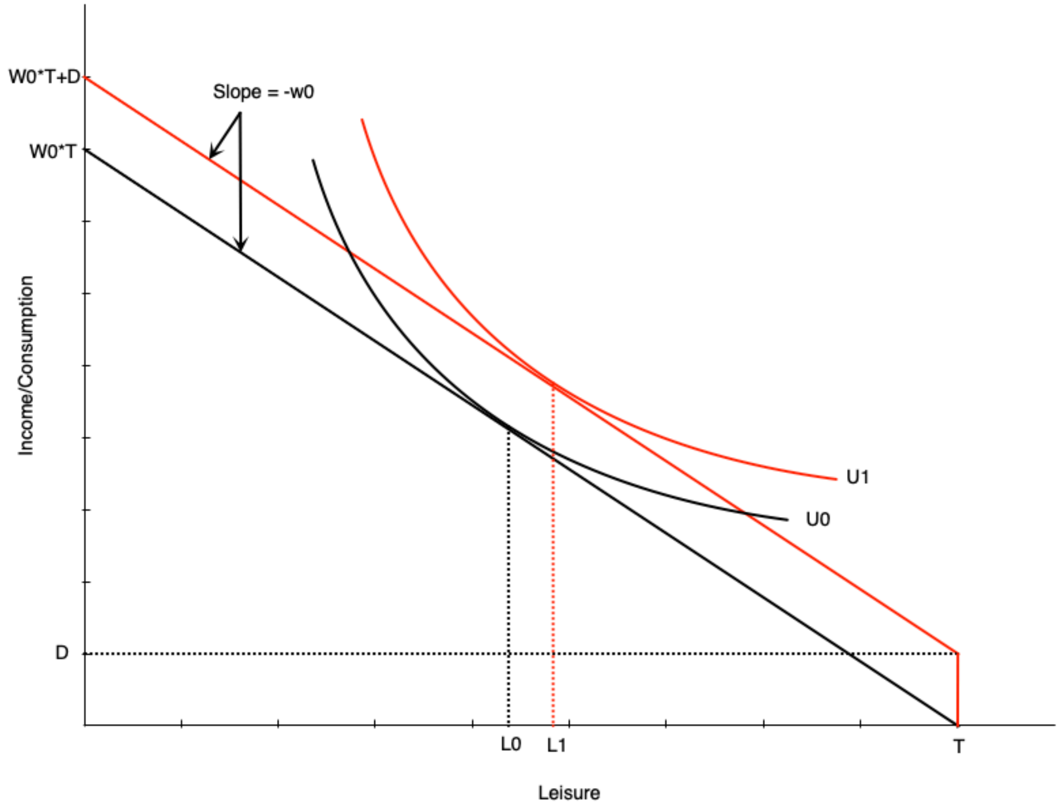
- Some programs reduce incentive to work
- Example: welfare benefits can encourage labour force dropout for those with strong preference for leisure

Demogrants, Welfare, Wage Subsidies

Four Standard Income Support Programs

We will examine four policy approaches:

Demogrant: Theory



How it works:

- Unconditional lump sum transfer D
- Shifts budget line up parallel
- Wage unchanged → **pure income effect**

Labor supply impact:

- If leisure is normal good → optimal leisure rises
- **Unambiguously negative work incentive**
- Reduces hours worked
- May induce non-participation

Demogrant: Canadian Example

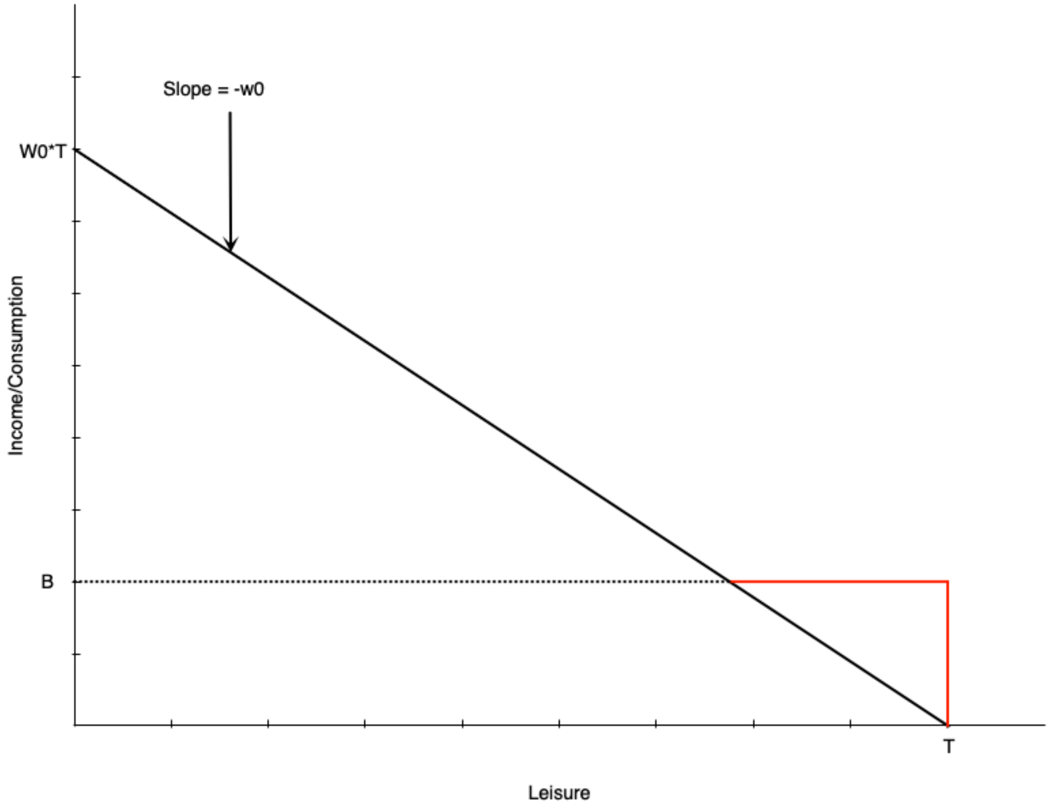
Universal Child Care Benefit (UCCB)

Welfare: Overview

Welfare = Social/Income Assistance in Canada

- Lump-sum payment to non-workers
- Reduced **dollar-for-dollar** with earned income

Welfare: Budget Constraint



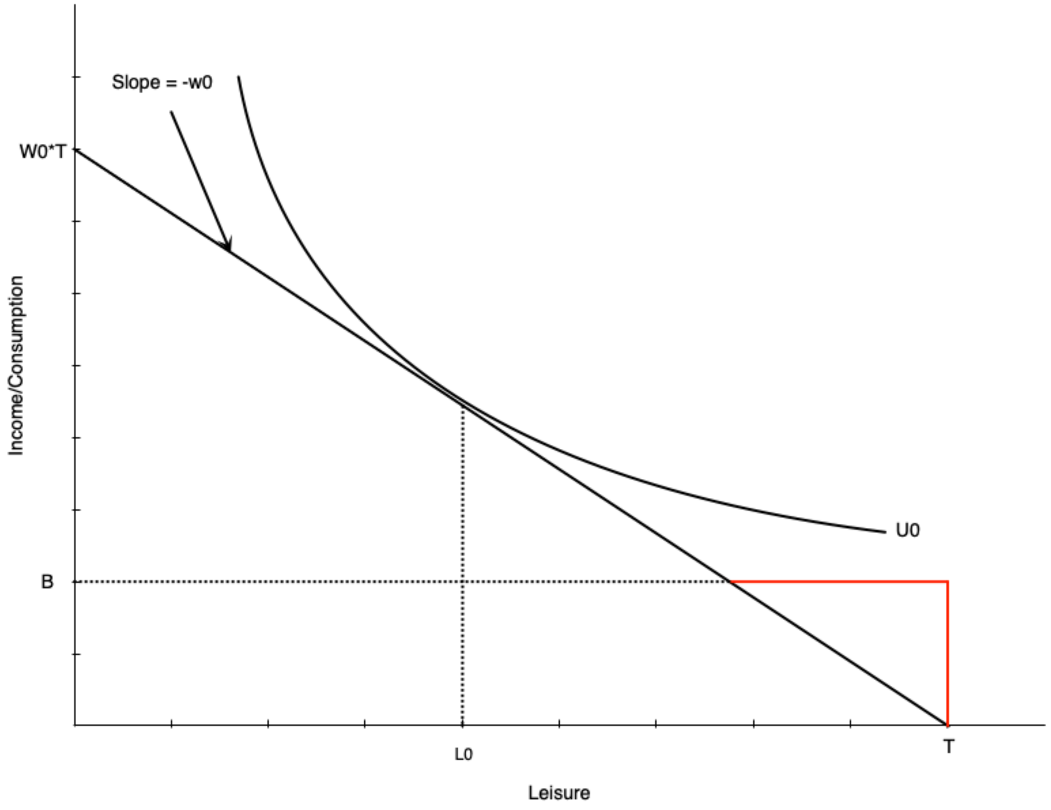
Budget line structure:

- **Black line:** original budget (leisure at L_0)
- **Red then black:** with welfare benefit B

Key features:

- Benefit B at zero hours of work
- Reduced \$1 for each \$1 earned
  - Equivalent to **100% tax rate** on earnings
- Income constant at B until earned income equals benefit
- After that, earn wage w_0 per hour worked

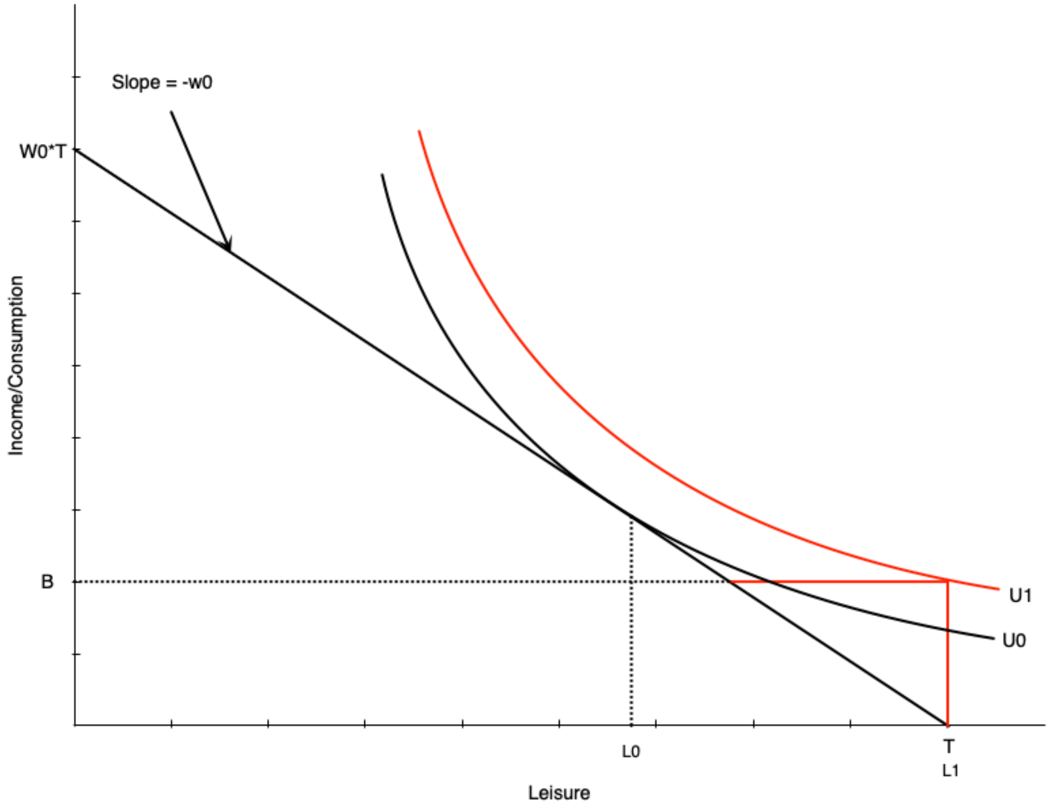
Welfare: No Effect Case



For some workers, welfare has no effect:

- Person working H_0 hours before benefit
- Still maximizes utility at H_0 hours after
- Generally occurs with people already working many hours

Welfare: Work Disincentive Case



Others face strong work disincentive:

- Person chooses **zero hours** after welfare benefit

This occurs when:

- Welfare benefit is generous
- Individual places high value on leisure

This is the problematic effect of welfare programs

Combating Work Disincentives

Policy options:

1. **Reduce benefit level**
    - But may not provide adequate support
2. **Increase wages**
    - Through minimum wages, subsidies, training, unions
    - Can be expensive and may reduce labour demand
3. **Change preferences**
    - Alter attitudes toward work/leisure
    - Difficult to achieve
4. **Negative income tax**
    - Reduce tax-back rate on earnings
    - Discussed next…

Welfare: Evidence from Quebec

Natural experiment: Welfare benefit structure changed in 1989

Before 1989:

- Age-based benefits for singles/couples with no kids
- Under 30: $185/month
- 30 and over: $507/month

Results at age 30:

- Much longer time on welfare
- Lower employment rates

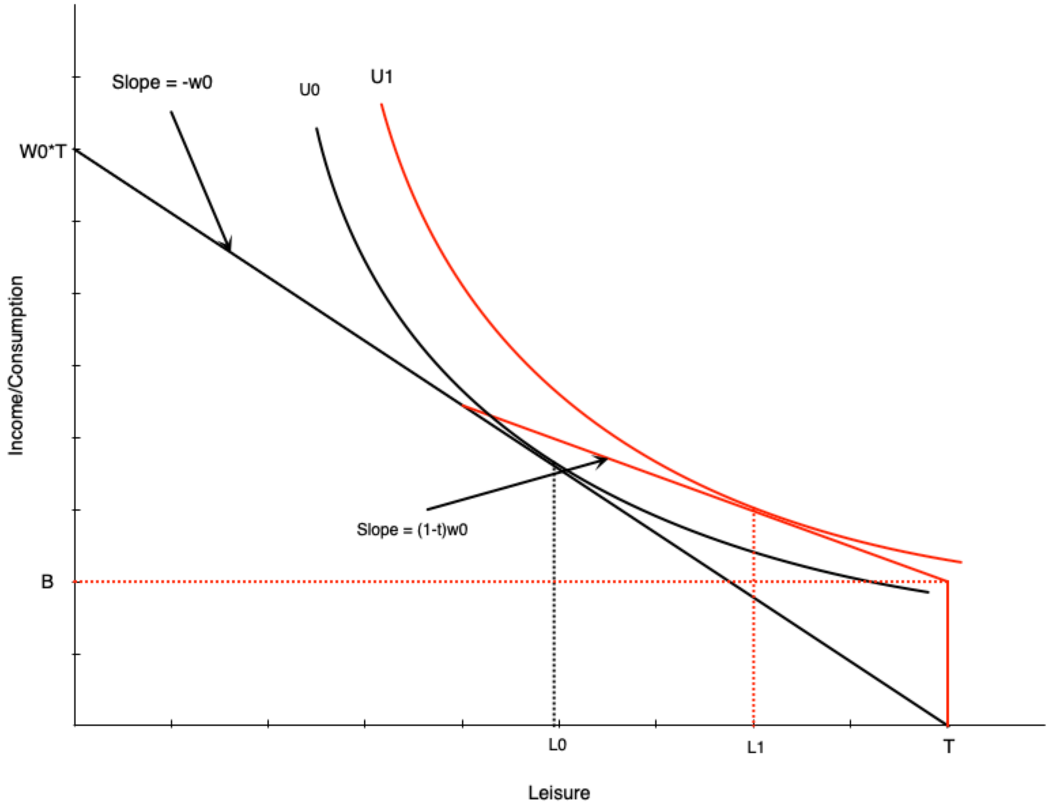
Turning 30 created a **large discontinuous jump** in benefits

Negative Income Tax: Overview

What is NIT?

- Welfare program where benefit reduced at rate **< 100%** of earnings
- Example: $10,000 benefit reduced by 50¢ per dollar earned

Negative Income Tax: Budget Constraint



NIT has two components:

- Benefit B at zero hours of work
- Tax rate t on earned income (until earned income equals benefit)

Effect on labour supply:

- Pivots budget line up vs. pure welfare
- Work hours reduced from H_0 to H_1
- **Work disincentive less than pure welfare**
- Person still works some hours (vs. dropping out with pure welfare)

Negative Income Tax: Income and Substitution Effects

Income effects:

Negative Income Tax: Math

The budget constraint with NIT:

$$C = \begin{cases} wh + B - twh, & \text{if } B - twh \geq 0 \\ wh, & \text{otherwise} \end{cases}$$

MINCOME: Manitoba’s NIT Experiment

Program details (1970s):

MINCOME Treatment Groups

Guarantee at enrolment	Tax Rate (on total Income)		
	35%	50%	75%
$3800	Plan 1	Plan 3	*
$4800	Plan 2	Plan 4	Plan 7
$5480	X	Plan 5	Plan 8

Plan 9

* Plan 6 was collapsed into Plan 7 because participants viewed this option as too punitive and left the experiment.
X This option was deemed to expensive for the budget.
Plan 9 was the control group

Source: Wikipedia

MINCOME: Results

Labour supply effects:

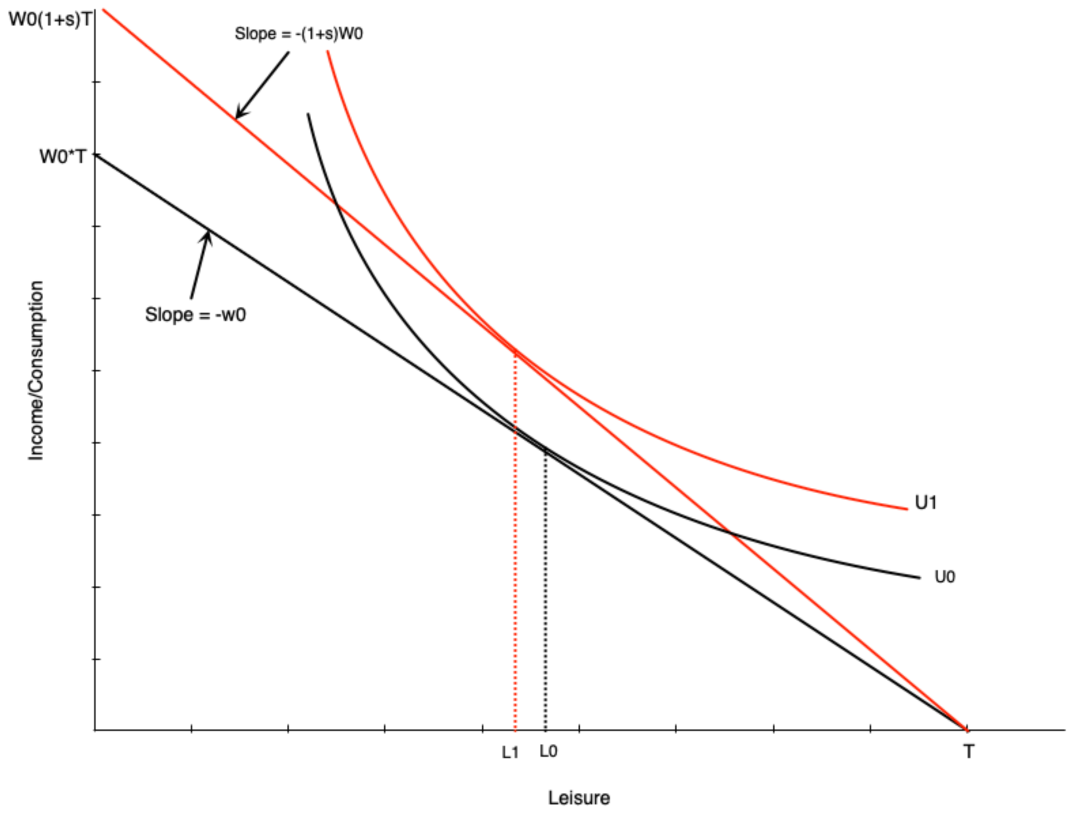
- Small decline in hours worked
  - Mainly mothers of newborns
  - Women in two-parent families
  - No significant decline for men

Wage Subsidies: Theory

What is a wage subsidy?

- Payment to workers that increases their wage rate
- Example: $5/hour subsidy increases $15/hour wage to $20/hour

Wage Subsidies: Diagram



Budget line pivots up with subsidy

- Graph shows person working more
- But effect is theoretically ambiguous
- Income and substitution effects work in opposite directions

Wage Subsidies: Policy Implications

Advantages:

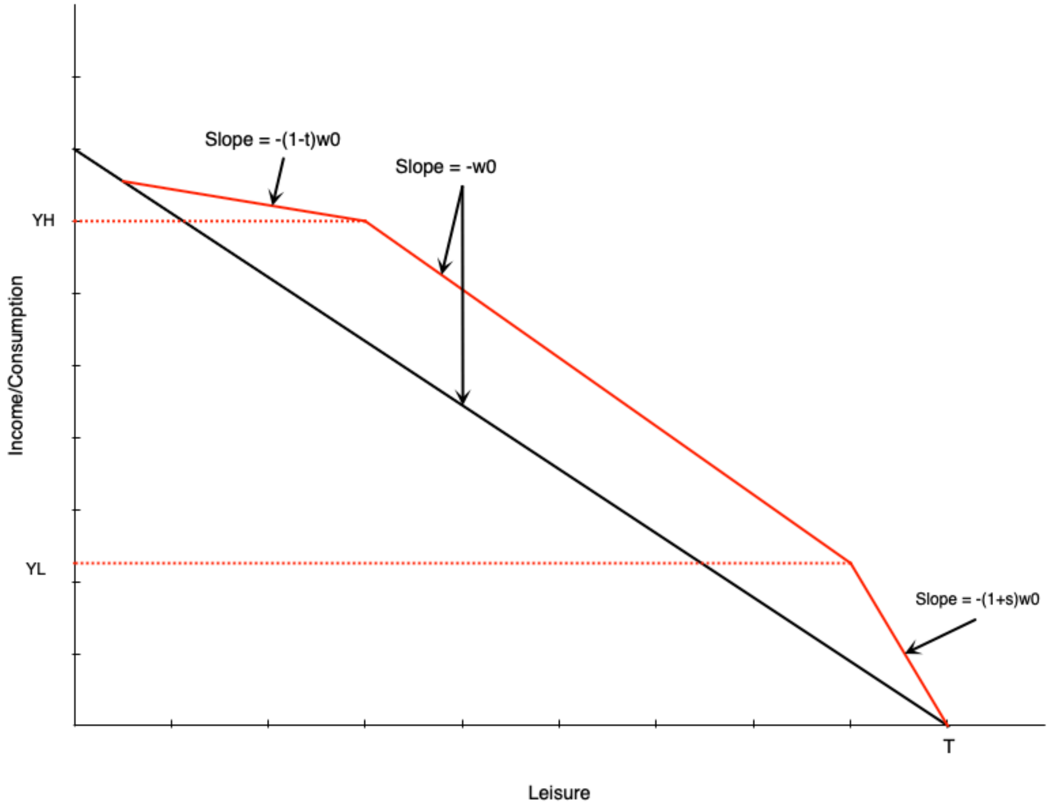
- Less negative work incentives than welfare or NIT
- No reduction in effective wage rate

Tax Credits: Overview

Two types of tax credits:

1. **Non-refundable:** Can reduce tax owed to zero (but not below)
2. **Refundable:** Can reduce tax owed below zero (results in payment)

Tax Credits: Budget Constraint



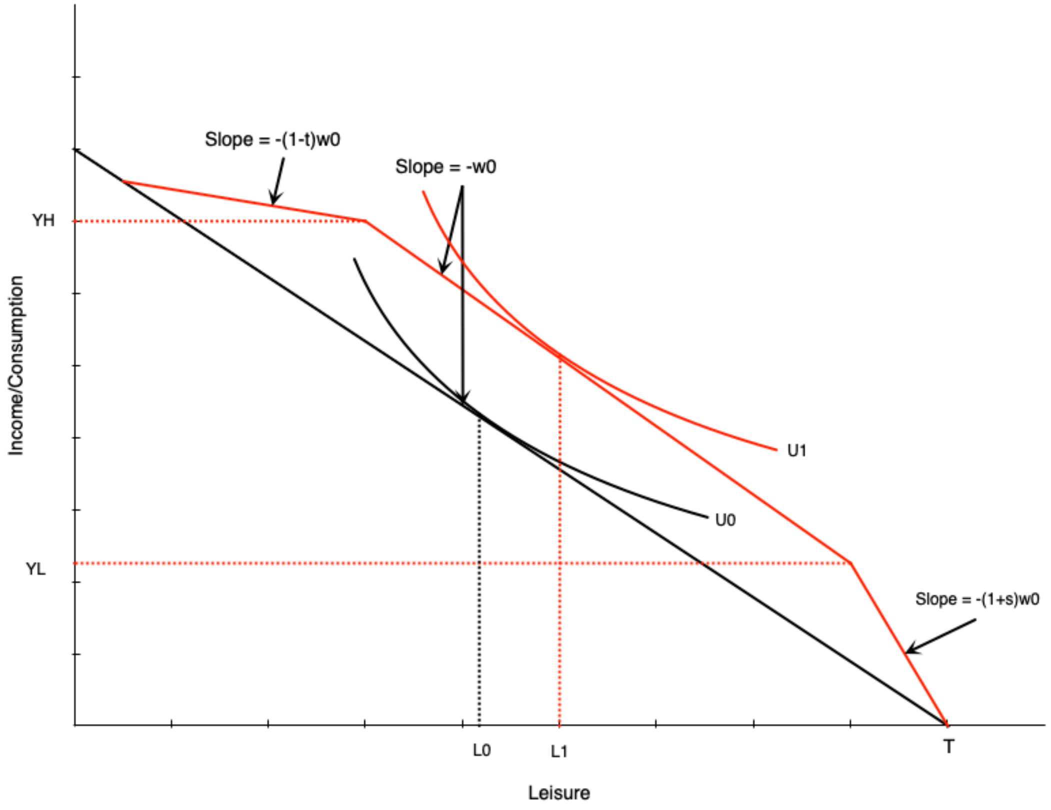
Three segments:

1. **Phase-in:** Subsidy proportional to earned income
2. **Plateau:** Maximum benefit level
3. **Phase-out:** Benefit reduced with earned income

Combines elements of:

- Wage subsidy (phase-in)
- Demogrant (plateau)
- NIT (phase-out)

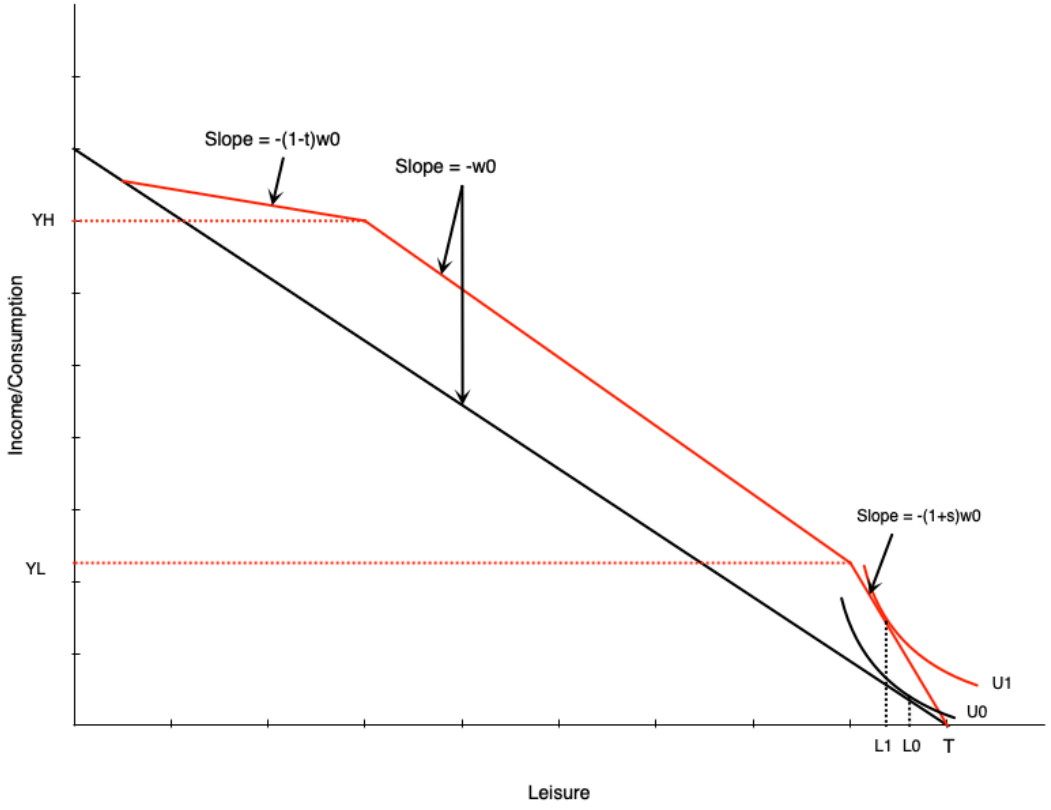
Tax Credits: Effect in Plateau Range



Person initially in middle range:

- Tax credit gives maximum benefit
- Acts like **pure income effect**
- If leisure is normal good → reduce work hours

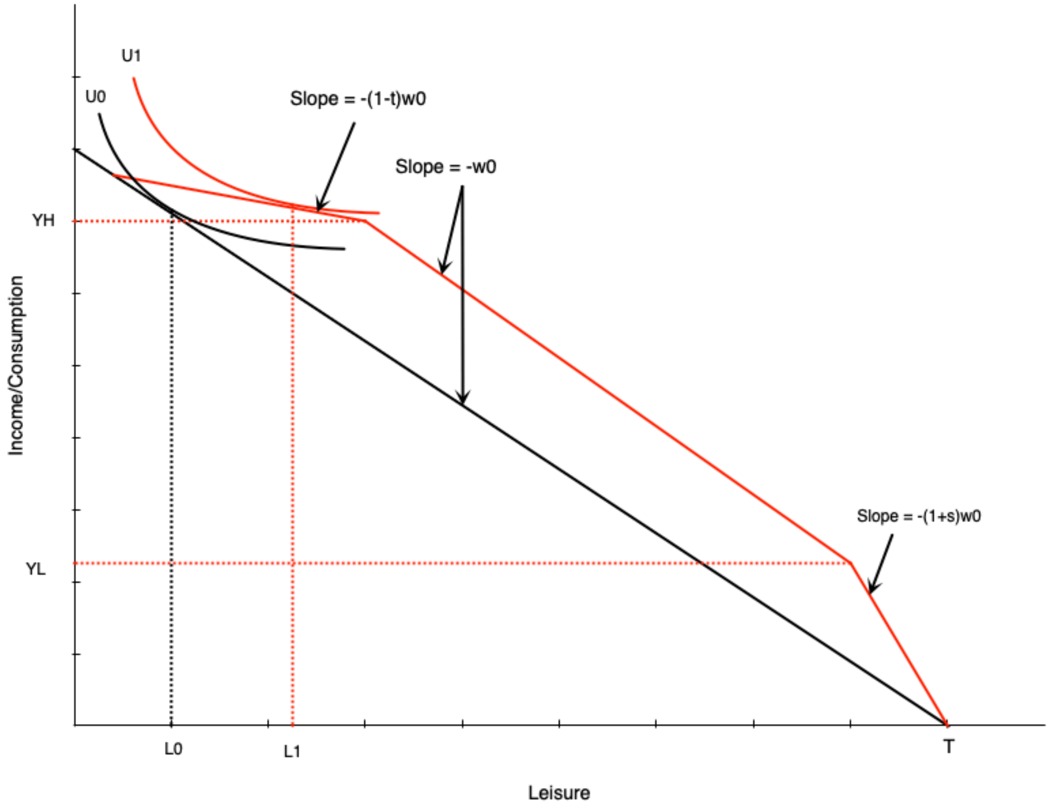
Tax Credits: Effect in Phase-In Range



Person in phase-in range:

- Sees increase in effective wage
- Acts like **pure wage subsidy**
- Income and substitution effects in opposite directions
- **Effect on hours is ambiguous**

Tax Credits: Effect in Phase-Out Range



Person in phase-out range:

- Similar to **negative income tax**
- Income effect → reduce hours ↓
- Substitution effect → reduce hours ↓
- **Effect on work is unambiguously negative**

Unemployment Insurance

Employment Insurance: Overview

Purpose:

- Safety net for workers who lose their job
- Collect income while unemployed

Employment Insurance: Benefits

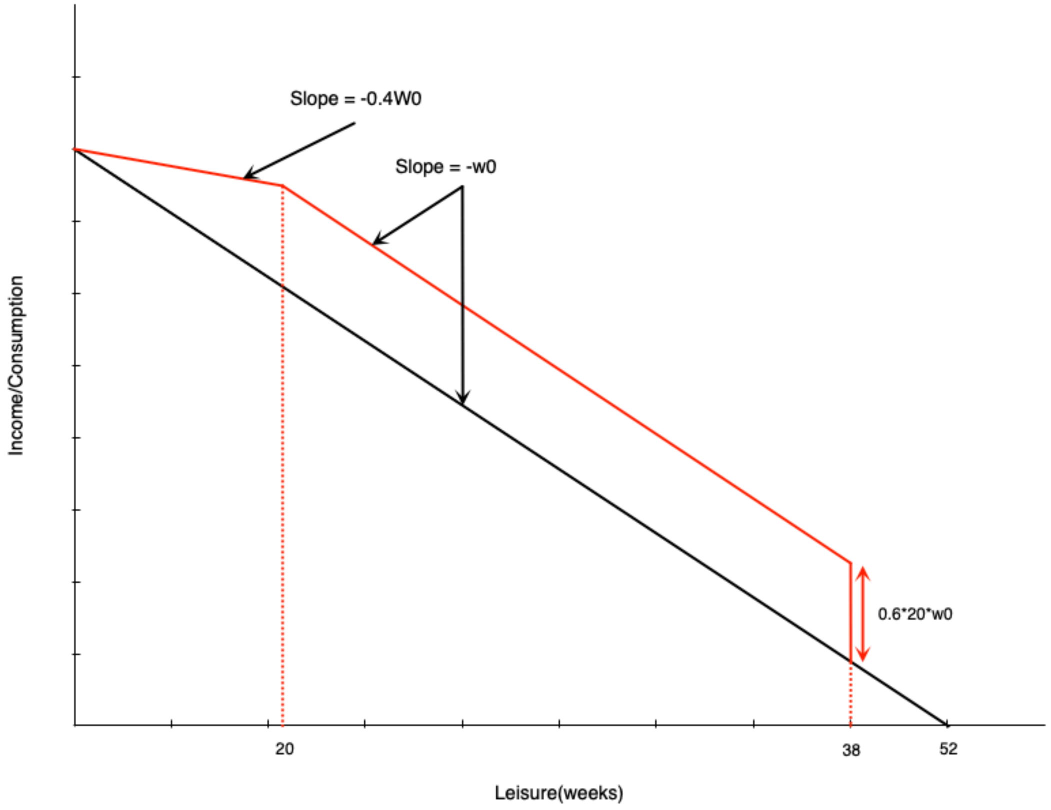
Regular benefits:

- 55% of average insurable weekly earnings
- Up to a maximum amount
- Duration: **14 to 45 weeks**
  - Depends on unemployment rate and hours worked in last year

EI: Simplified Model Parameters

To analyze with income-leisure diagram:

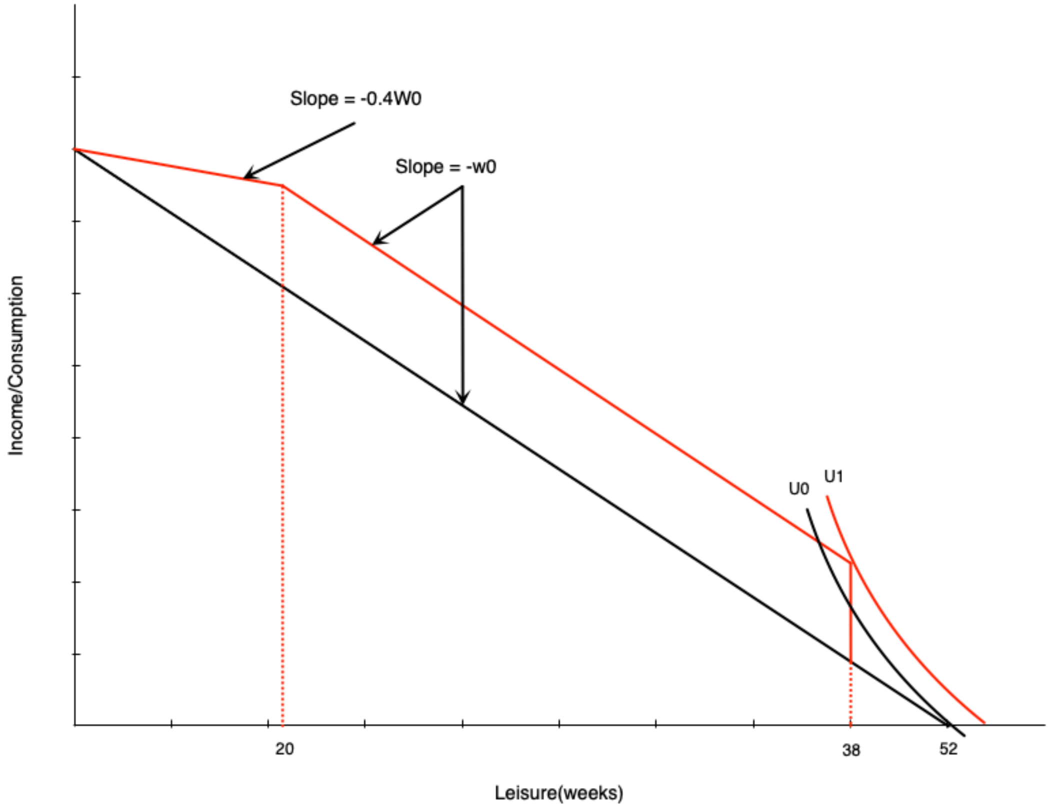
EI: Budget Constraint



Three segments:

1. **First 14 weeks:** Paid at usual wage w_0
2. **Weeks 14-32 of work (20-38 weeks leisure):** Get 20 weeks of benefits
   - Benefit = $0.6 \times 20 \times w_0$
3. **More than 32 weeks of work (<20 weeks leisure):** Benefits reduced week-for-week
   - Benefits paid at 60% of wage
   - Work paid at 100% of wage
   - **Keep 40% of wage** in this range

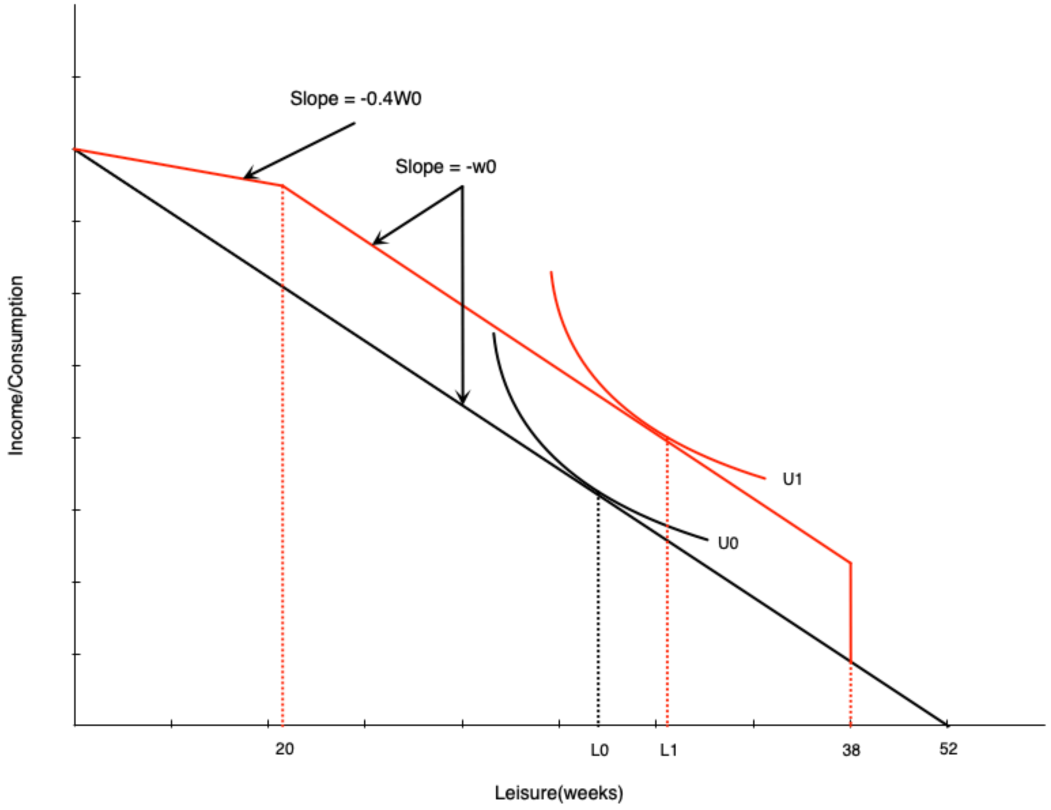
EI: Effect on Non-Participants



A non-participant might start working:

- Large increase in non-labour income after 14 hours of work
- Induces labour force entry

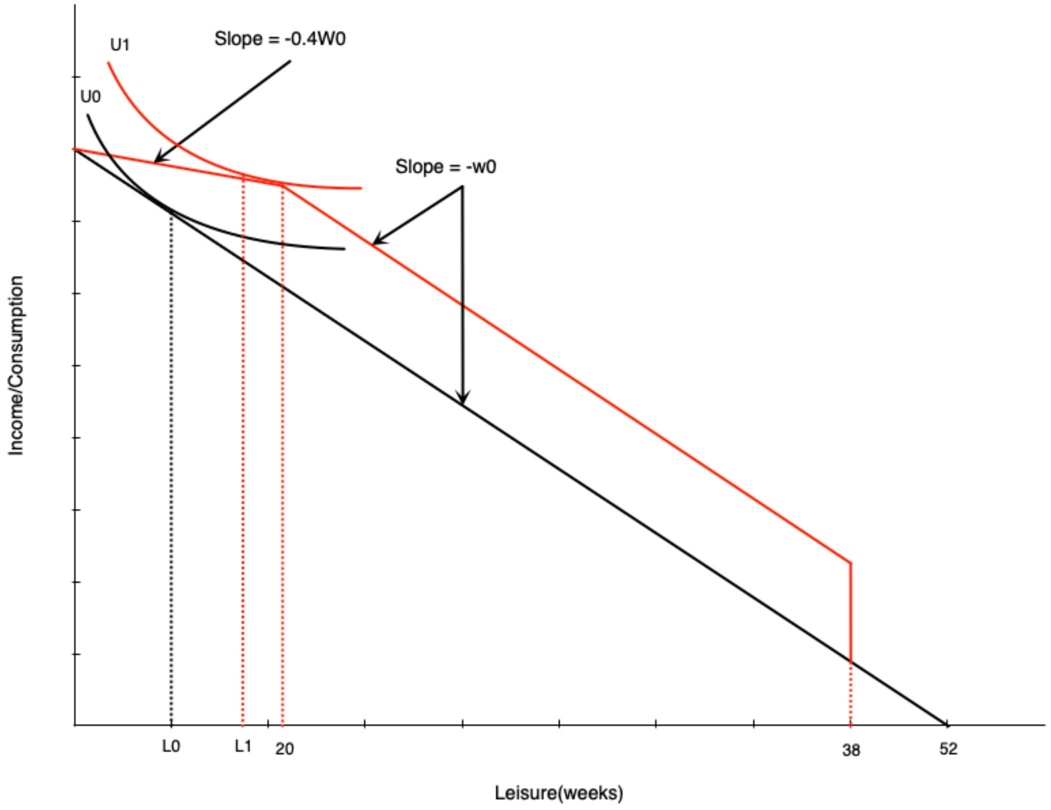
EI: Effect in Middle Range



Person getting full benefits:

- Acts like **pure income effect**
- Will reduce work weeks

EI: Effect on High-Hour Workers



In phase-out range:

- Acts like **negative income tax**
- Income effect → reduce hours ↓
- Substitution effect → reduce hours ↓
- **Effect unambiguously negative**

Evidence: Maine vs. New Brunswick

Natural experiment:

Evidence: Maine vs. New Brunswick Results

Analysis findings:

New Brunswick experienced:

- Higher labour force participation for marginally-attached workers
- Lower hours for those with stronger attachments

Interpretation:

- Corresponds exactly to model predictions
- Highlights work disincentive effects of income maintenance programs

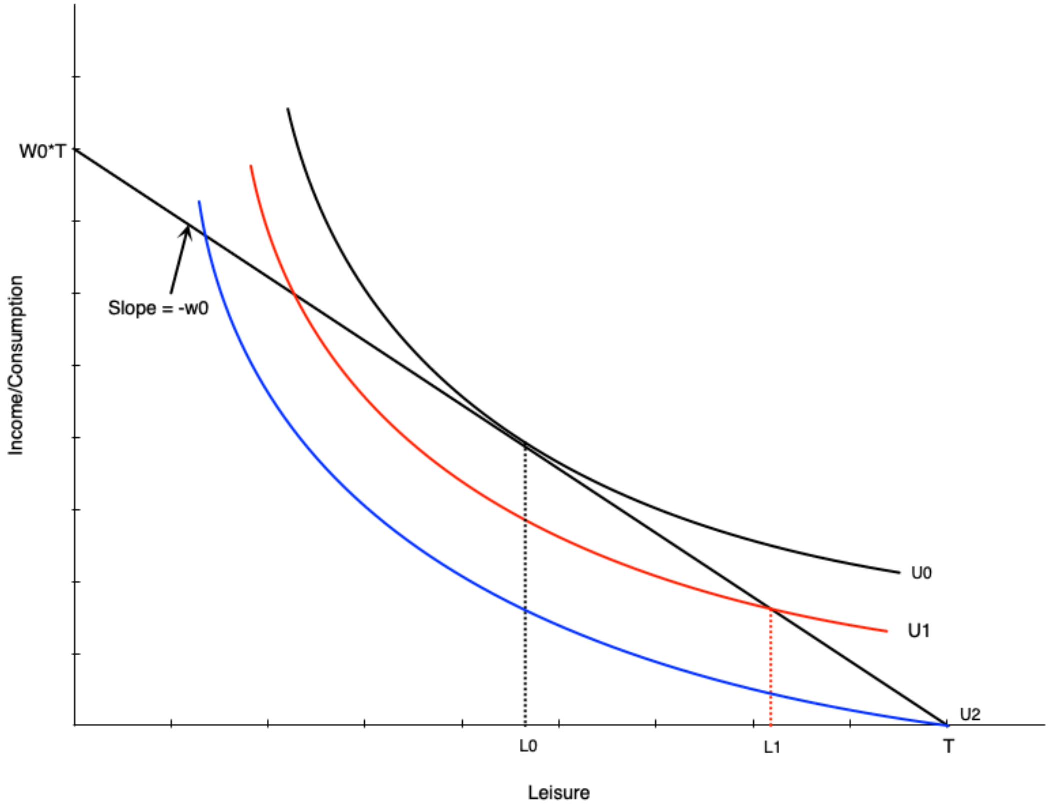
Disability and Workers Compensation

Introduction

What are these programs?

- Provide income support to injured/ill workers
- Come from public and private sources
- Premiums paid by employers and employees

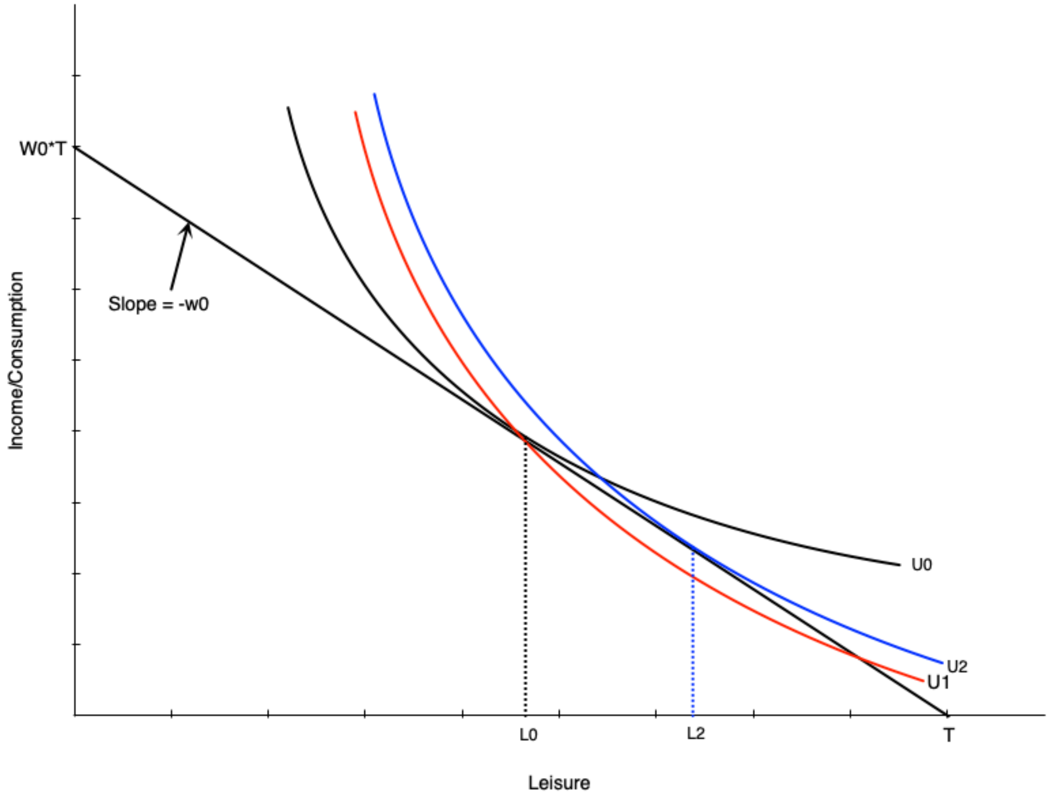
Effect of Disability on Labour Supply



Three scenarios:

1. **Initially:** Choose L_0 leisure, work H_0 hours
2. **Injured but can work:** Choose L_1 leisure, work H_1 hours
   - Lower (non-optimal) indifference curve
3. **Completely disabled:** Choose T leisure, zero work
   - Even lower indifference curve

Disability and Preferences



Disability may change preferences:

- Work becomes more difficult
- Preferences tilt toward leisure
- Indifference curve becomes steeper
- MRS increases at each point
- More willing to give up consumption for leisure

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Result:

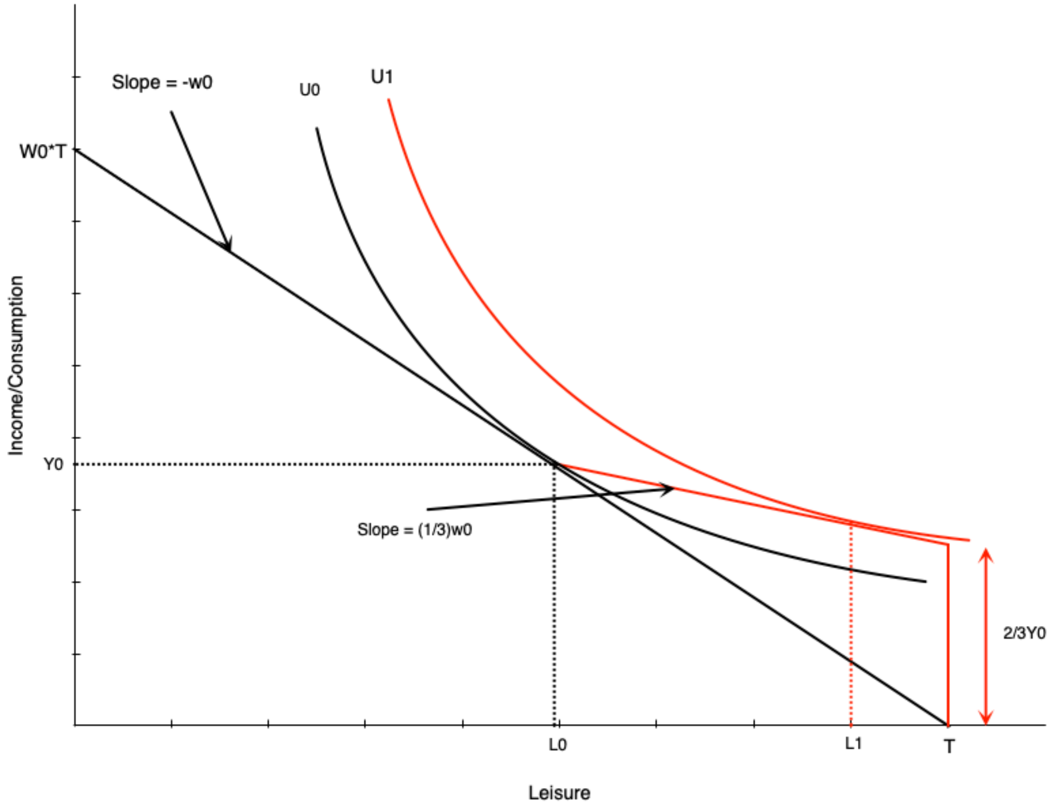
- Even if unconstrained, choose more leisure
- Person moves from L_0 to L_2

Compensation Schemes

Typical disability insurance structure:

- Pays a fraction of previous earnings
- **Example (Ontario WSIB):**
  - 85% of net earnings (up to maximum)
  - Lasts until return to work or age 65

Two-Thirds Compensation



Looks like negative income tax:

- At zero hours: get 2/3 of previous income
- For each hour worked:
  - Gain full wage
  - Lose 2/3 of wage in benefits
  - **Keep 1/3 of wage**

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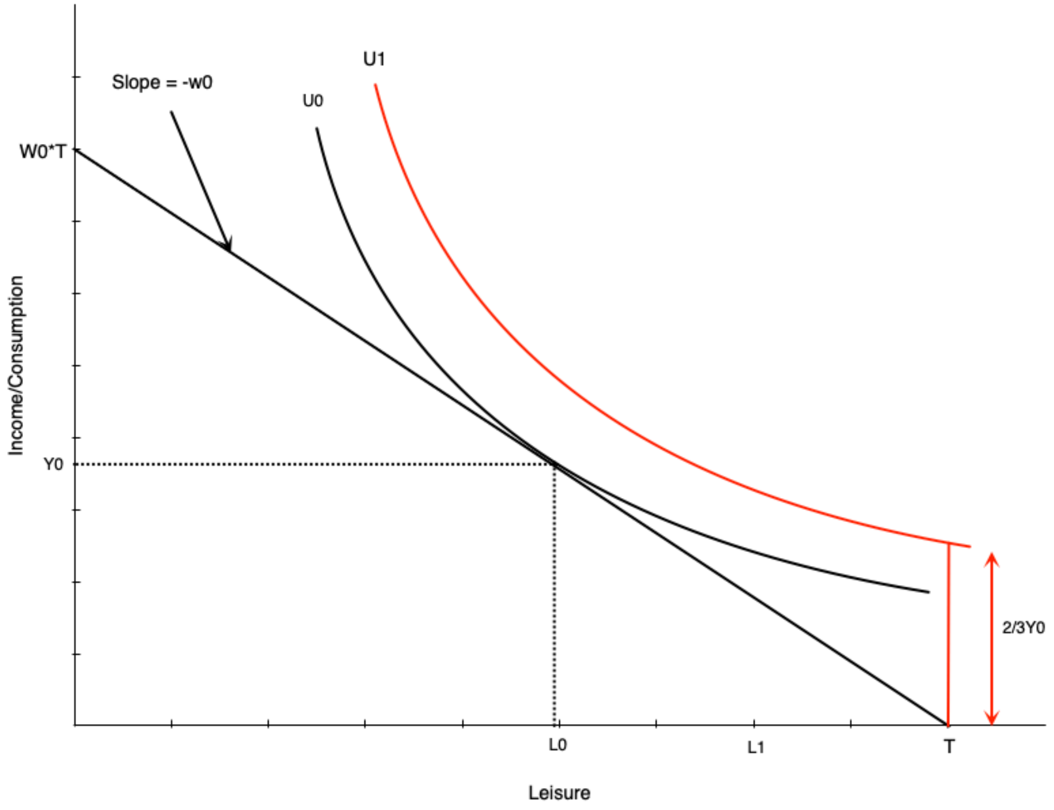
For workers with flexible hours:

- Effects match NIT
- Negative effect on hours worked

Restricted Hours: Generous Compensation



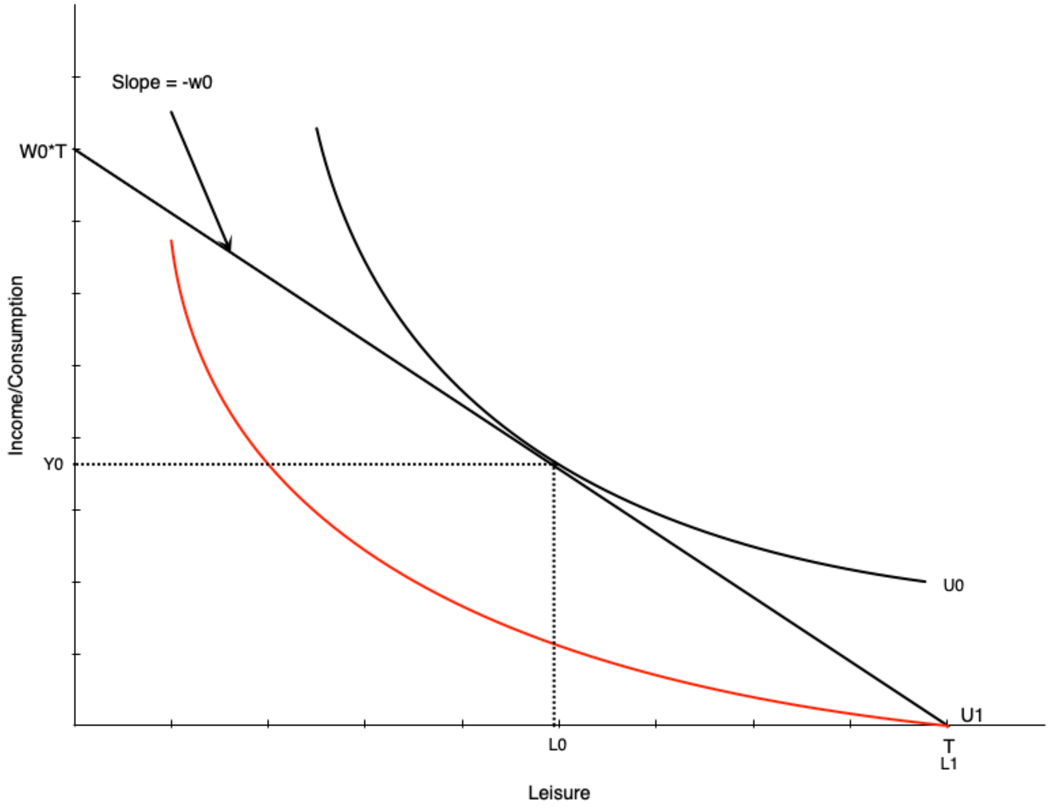
What if worker can only work H_0 hours or zero?

- If scheme generous enough → choose zero hours
- Utility at zero hours > utility at H_0 hours

Implication:

Strong work disincentives of generous scheme

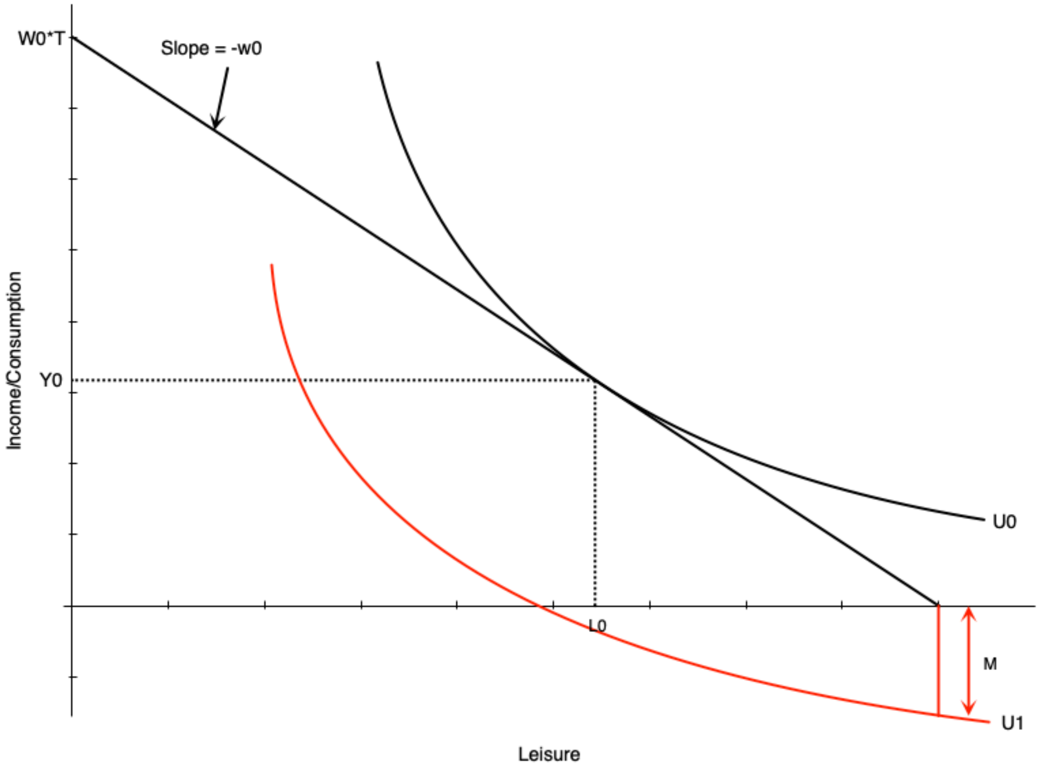
No Compensation Case



Not providing compensation is problematic:

- Person who can work chooses H_0 hours
- Person who cannot work gets lower utility
- End up substantially worse off

Medical Expenses Case

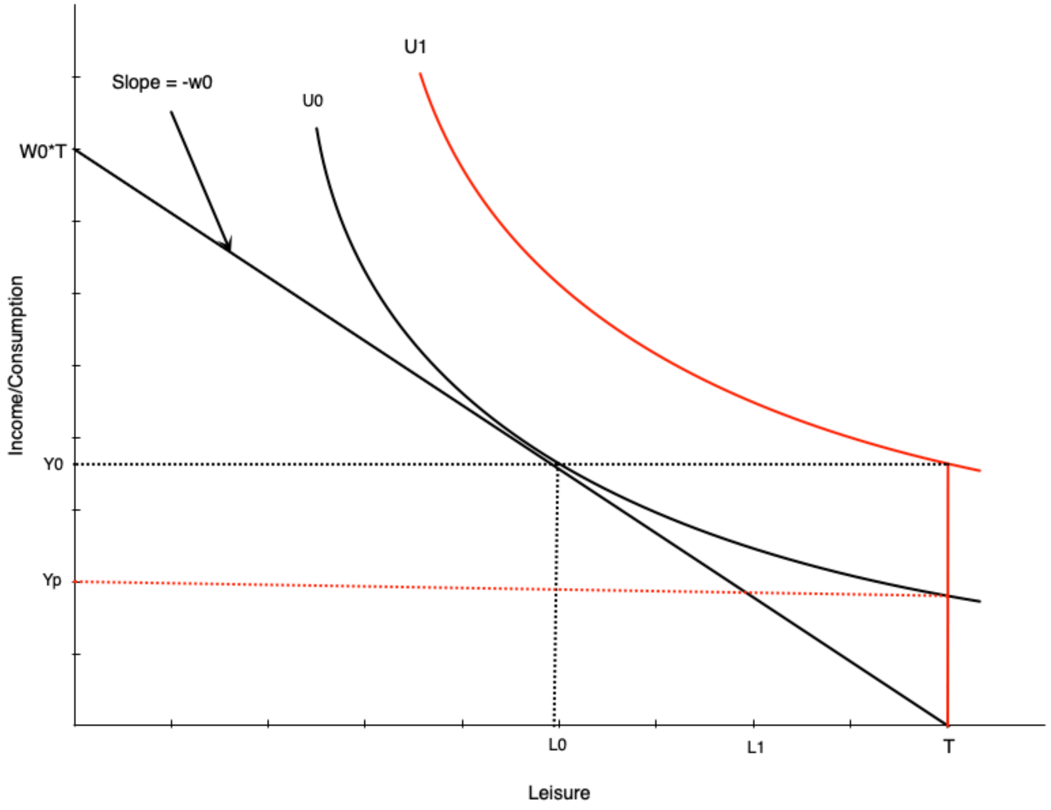


Medical expenses worsen situation:

- Act like reduction in non-labour income
- If large enough, non-labour income could go negative
- Occurs with no/very low compensation

This is an extreme scenario

Optimal Compensation Level



How much to compensate non-workers?

Two options:

1. **Full income replacement**
   - Higher indifference curve (U_1) than working
2. **Same satisfaction as working**
   - Same indifference curve as working
   - Lower compensation (Y_p)

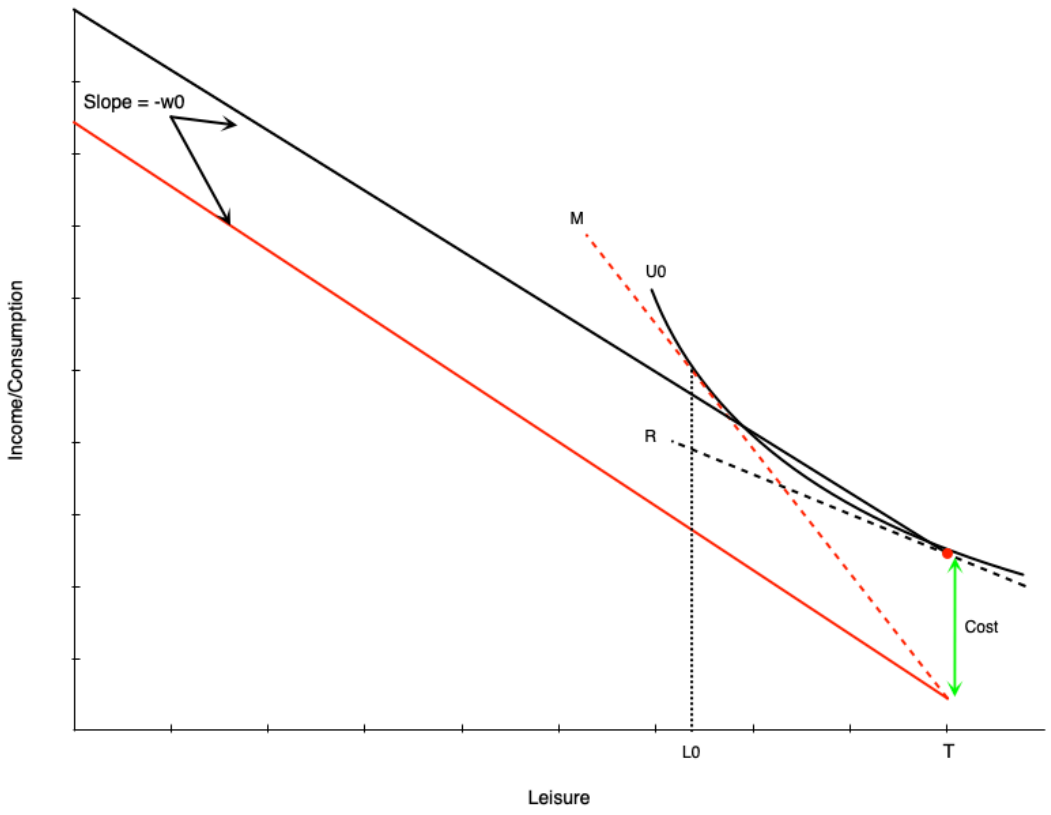
The optimal level is a policy question

Child Care Subsidies

Introduction

Canada's child care expansion:

Child Care Costs: Non-Working Case



Budget line becomes discontinuous:

- At zero hours: red dot
- For any work: drops by child care costs
- Then straight line with slope = wage

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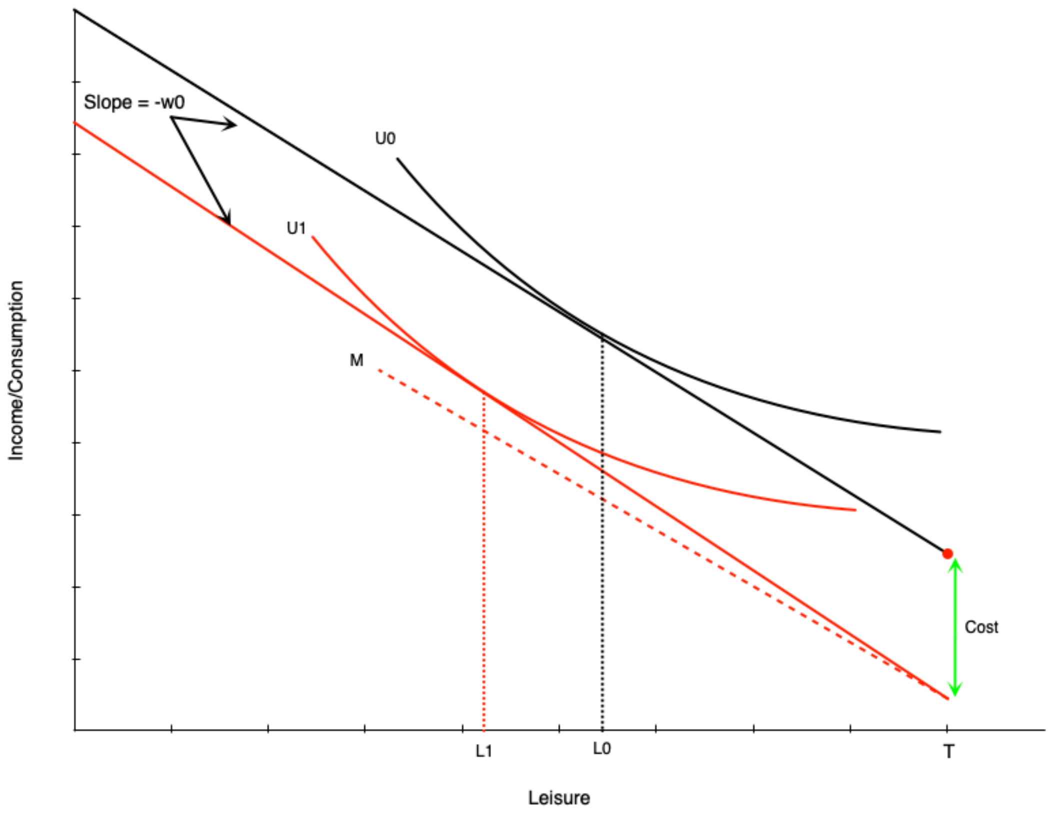
Effect:

- Child care costs **increase reservation wage**
- Need higher wage to induce work
- This person chooses not to work

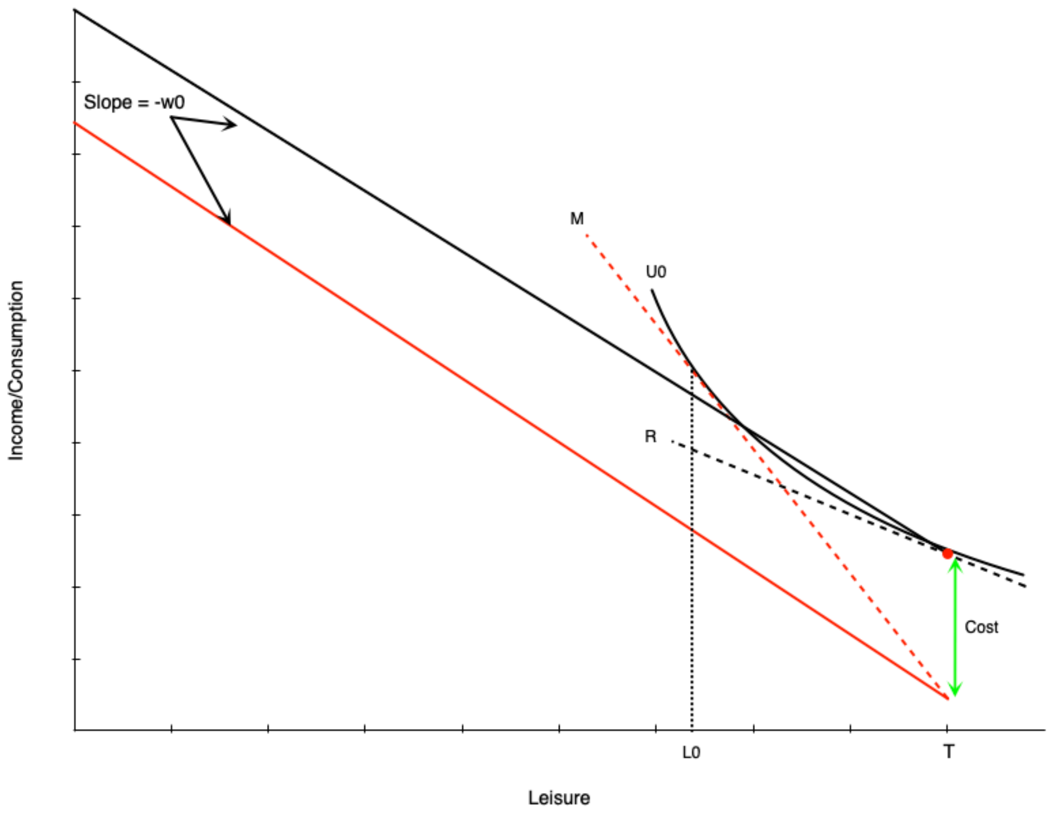
Child Care Costs: Working Case



Person with reservation wage above wage + costs:

- Works positive hours
- Child care costs = **negative pure income effect**
- Choose less leisure, more work
- Must work more to offset income loss

Child Care Costs: Labour Supply Gap



Creates gap in labour supply schedule:

- Person who works will work minimum of $H_0 = T - L_0$ hours
- Not worth working less due to large child care costs

Result:

Labour supply schedule jumps from zero to H_0 hours

Child Care Subsidies: Effects

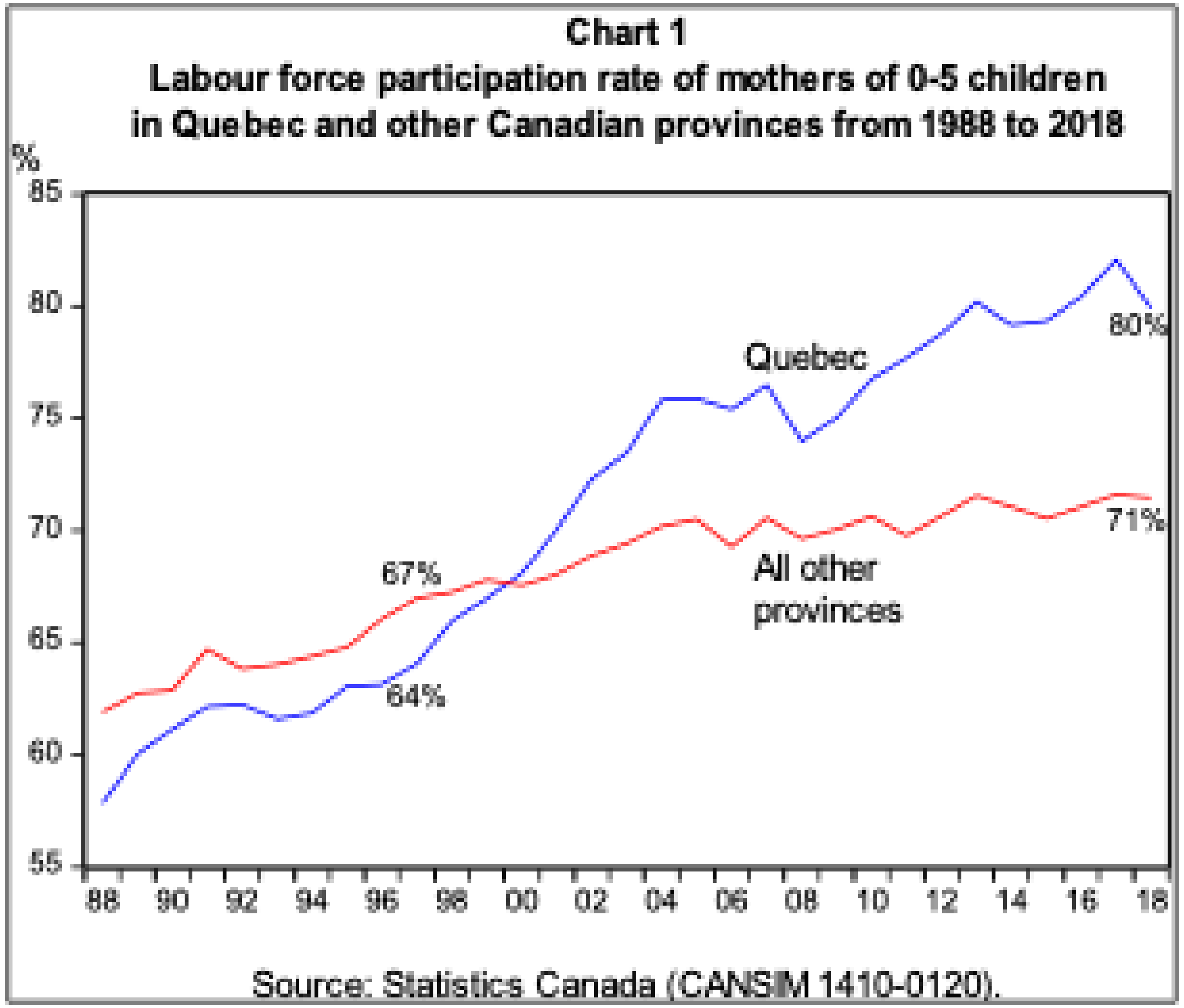
A full subsidy (costs → zero) would:

Evidence: Quebec's $5/Day Program

Program details:

- Implemented in 2000
- Currently sliding scale: $8.25 to $21.45

Evidence: Visual Summary



Source: Fortin (2019)

Summary

Key Takeaways

Income maintenance programs: